

PROVISIONAL PATENT SPECIFICATION

Autonomous Compositional Reasoning Architecture Integrated with Persistent Existence Frame (Aurora + PEF)

INVENTION TITLE

Autonomous Compositional Reasoning Architecture Integrated with Persistent Existence Frame (Aurora + PEF)

RELATED APPLICATIONS

This application is a continuation-in-part and improvement of Australian Provisional Patent Application No. 2025905835, titled "Aurora - Autonomous Conceptual Reasoning Architecture," filed 27 November 2025.

This application incorporates and builds upon Australian Provisional Patent Application No. 2025905860, titled "Systems and Methods for Implementing a Persistent Existence Frame (PEF) for Non-Temporal Cognitive and Machine Reasoning Architectures," filed 28 November 2025.

This application describes the integration and implementation of Aurora's compositional reasoning primitives operating within the Persistent Existence Frame substrate, and specifies the detailed architecture, processing mechanisms, and explicit primitives that implement the conceptual framework described in the earlier Aurora filing.

ABSTRACT

This invention provides an autonomous compositional reasoning architecture that generates, transforms, and evaluates conceptual structures using explicit primitives and structured state representations. The system comprises roles, domains, and spans operated upon by formal compositional primitives (including WE, THEN, WHILE, UNTIL, and BECAUSE), each defined by preconditions, activation rules, update logic, and collapse conditions. A processing architecture ("Aurora Block") interprets the conceptual state, evaluates primitive eligibility, enforces constraints, manages spans, and synthesizes role transformations to produce coherent reasoning outputs. In certain embodiments, the architecture operates within a Persistent Existence Frame (PEF), a non-temporal existential substrate enabling continuous-state reasoning without temporal encoding. The invention supports deterministic, interpretable reasoning independent of statistical training or transformer mechanisms, enabling structured conceptual computation across diverse implementations. This specification provides detailed implementation of the Aurora architecture previously filed, specifying explicit compositional primitives (WE, THEN, WHILE,

UNTIL, BECAUSE), structured state representations (Roles, Domains, Spans), and processing architecture (Aurora Block), integrated with the Persistent Existence Frame as described in related application 2025905860.

FIELD OF THE INVENTION

The invention relates to artificial intelligence, machine reasoning, and cognitive architectures.

Specifically, it concerns systems that generate and transform conceptual structures using explicit compositional operators within structured conceptual states comprising roles, domains, and spans.

In certain embodiments, the architecture may operate within a non-temporal existential substrate.

BACKGROUND OF THE INVENTION

Transformer-based AI systems model language using statistical prediction across token sequences. While effective for surface-level fluency, they exhibit fundamental limitations:

1. **Premature ambiguity collapse** — Transformers select a single continuation even when multiple interpretations remain valid.
2. **Absence of explicit compositional operators** — They do not contain reasoning primitives such as WE, THEN, WHILE, UNTIL, or BECAUSE.
3. **Dependence on temporal encoding** — Transformers require positional encodings or recurrence to model sequence order.
4. **Lack of explicit structural state** — Existing models do not maintain persistent roles, domains, or spans.
5. **Opacity and non-determinism** — Reasoning behaviour is emergent from training data, not explicit mechanism.

No existing architecture performs reasoning by applying explicit compositional primitives within a structured conceptual state, nor within a non-temporal existential framework such as a Persistent Existence Frame (PEF).

This invention introduces a fundamentally different approach.

SUMMARY OF THE INVENTION

The invention provides a reasoning architecture composed of:

1. A Conceptual State Representation

Including:

- **Roles:** entity representations with stable identity
- **Domains:** verb-driven activation contexts
- **Spans:** bounded conceptual episodes

2. Compositional Primitives

Explicit reasoning operators including:

- WE (collective entity formation)
- THEN (conceptual progression)
- WHILE (conditional continuity)
- UNTIL (goal-state termination)
- BECAUSE (causal attribution)

Each primitive includes defined preconditions, activation logic, update rules, and collapse conditions.

3. Aurora Block Processing Architecture

A deterministic pipeline comprising:

- State Interpreter
- Primitive Evaluator
- Constraint Engine
- Domain Selector
- Span Manager
- Role Synthesizer
- Collapse Resolver
- Output Generator

4. Optional PEF Integration

In some embodiments, reasoning occurs within a Persistent Existence Frame (PEF) providing non-temporal execution, reconstruction, and projection.

This architecture enables explicit, interpretable reasoning without statistical training or temporal sequence modeling.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 — High-Level System Architecture

FIG. 2 — Conceptual State Structure (Roles, Domains, Spans)

FIG. 3 — Primitive Operation Flow

FIG. 4 — WE Primitive Example

FIG. 5 — THEN Primitive Linking Two Spans

FIG. 6 — WHILE Conditional Span

FIG. 7 — UNTIL Termination Structure

FIG. 8 — BECAUSE Causal Linkage

FIG. 9 — Aurora Block Processing Components

FIG. 10 — Optional Integration with Persistent Existence Frame

DETAILED DESCRIPTION OF THE INVENTION

5.1 Overview

The architecture performs conceptual reasoning by applying explicit compositional primitives to a structured conceptual state.

The system maintains predictable, interpretable transformations without relying on learned statistical patterns, temporal sequences, or positional encodings.

Aurora operates with or without a Persistent Existence Frame.

When PEF is present, reasoning occurs in a non-temporal existential substrate.

5.2 Persistent Existence Frame (Optional)

In certain embodiments, the architecture operates within a **Persistent Existence Frame (PEF)**, as defined in Australian Provisional Patent Application No. 2025905860.

PEF provides:

- a continuous present-state substrate
- reconstruction of prior states using echo-traces
- projection of potential conceptual states

PEF eliminates all need for temporal tracking or sequence encoding.

Aurora remains fully functional with or without PEF.

5.3 Conceptual State Representation

The system maintains:

5.3.1 Roles

Conceptual entities with persistent identity and attributes.

5.3.2 Domains

Verb-driven contexts that structure interactions between roles.

5.3.3 Spans

Bounded episodes containing conceptual transformations.

Roles enter domains; spans encapsulate transformations and collapse at defined conditions.

5.4 Conceptual State Dynamics

Aurora updates conceptual state by:

- Primitive activation
- Domain selection
- Span creation and collapse
- Role synthesis
- Constraint resolution

Ambiguity is preserved until logical resolution.

6. COMPOSITIONAL PRIMITIVES

The foundational primitives include:

6.1 WE Primitive

Purpose: To form a composite entity with shared agency.

Preconditions: mutual qualification, domain eligibility, non-conflict.

Activation: composite role creation; WE-span opened.

Update: attribute propagation; collective agency.

Collapse: divergence, domain requirement mismatch, or forced separation.

6.2 THEN Primitive

Purpose: Conceptual progression without temporal encoding.

Preconditions: prior span complete; role continuity; domain compatibility.

Activation: new span; continuity mapping.

Update: role propagation; ambiguity retention.

Collapse: end of sequence.

6.3 WHILE Primitive

Purpose: Maintain a structure under active condition.

Preconditions: a true condition domain.

Activation: persistent span.

Update: internal modifications under constraint.

Collapse: condition false or overriding primitive.

6.4 UNTIL Primitive

Purpose: Maintain activity until target condition achieved.

Preconditions: active domain + termination condition.

Activation: ongoing span with embedded target.

Update: structural continuity.

Collapse: target satisfied.

6.5 BECAUSE Primitive

Purpose: Causal linkage between spans.

Preconditions: completed cause-span + effect-span requiring explanation.

Activation: causal mapping.

Update: attribute propagation.

Collapse: causal closure.

7. AURORA BLOCK: PROCESSING ARCHITECTURE

The Aurora Block comprises:

1. **State Interpreter**
2. **Primitive Evaluator**
3. **Constraint Engine**
4. **Domain Selector**
5. **Span Manager**
6. **Role Synthesizer**
7. **Collapse Resolver**
8. **Output Generator**

Processing Cycle:

1. Interpret current state
2. Evaluate primitive eligibility
3. Validate transformations
4. Select domain
5. Manage spans
6. Synthesize roles
7. Resolve collapse
8. Generate new conceptual state

The architecture supports nested spans, parallel ambiguities, compositional chaining, and deterministic reasoning.

8. CLAIMS

Claim 1 (Independent Claim — System Claim):

A machine reasoning system comprising:

- (a) a conceptual state representation comprising roles, domains, and spans;
- (b) a set of explicit compositional primitives, each primitive defined by preconditions, activation rules, update logic, and collapse conditions; and
- (c) a processing architecture configured to:
 - (i) interpret the conceptual state,
 - (ii) evaluate primitive eligibility,
 - (iii) enforce conceptual constraints,

- (iv) activate primitives,
- (v) update roles, domains, and spans in accordance with said primitives, and
- (vi) generate a new conceptual state.

Primitive-Specific Claims

Claim 2:

The system of Claim 1, wherein the compositional primitives include a **WE primitive** that:

- (a) evaluates role compatibility,
- (b) creates a composite role when preconditions are satisfied,
- (c) maintains collective agency for the duration of a span, and
- (d) collapses into constituent roles when divergence or domain separation occurs.

Claim 3:

The system of Claim 1, wherein the compositional primitives include a **THEN primitive** that:

- (a) links a completed conceptual span to a subsequent span,
- (b) preserves role continuity across spans, and
- (c) establishes conceptual progression without temporal encoding.

Claim 4:

The system of Claim 1, wherein the compositional primitives include a **WHILE primitive** that:

- (a) initiates a persistent span governed by a condition domain,
- (b) maintains conceptual updates while the condition remains true, and
- (c) collapses when the condition becomes false or a terminating primitive activates.

Claim 5:

The system of Claim 1, wherein the compositional primitives include an **UNTIL primitive** that:

- (a) creates a span representing ongoing activity,
- (b) embeds a target condition defining span termination, and
- (c) collapses upon satisfaction of the target condition.

Claim 6:

The system of Claim 1, wherein the compositional primitives include a **BECAUSE**

primitive that:

- (a) establishes a causal relationship between a first span and a second span,
- (b) propagates explanatory structure from the first span to the second span, and
- (c) collapses when causal linkage is complete or superseded.

State Representation Claims

Claim 7:

The system of Claim 1, wherein roles represent conceptual entities possessing persistent identity and attributes.

Claim 8:

The system of Claim 1, wherein domains represent verb-activated contexts that structure interactions between roles.

Claim 9:

The system of Claim 1, wherein spans represent bounded conceptual episodes that encapsulate reasoning events and collapse according to primitive-specific termination rules.

Processing Architecture Claims

Claim 10:

The system of Claim 1, wherein the processing architecture comprises:

- (a) a state interpreter,
- (b) a primitive evaluator,
- (c) a constraint engine,
- (d) a domain selector,
- (e) a span manager,
- (f) a role synthesizer, and
- (g) a collapse resolver.

Claim 11:

The system of Claim 10, wherein the primitive evaluator activates primitives based on deterministic rule evaluation rather than statistical prediction.

Claim 12:

The system of Claim 10, wherein the constraint engine enforces role compatibility, domain validity, attribute inheritance, non-contradiction, and span integrity.

Claim 13:

The system of Claim 10, wherein the span manager supports nested spans, parallel spans, and ambiguity-preserving spans.

Claim 14:

The system of Claim 10, wherein the role synthesizer creates composite roles, splits roles upon collapse, and propagates attributes through spans.

PEF Integration Claims (Optional Substrate)**Claim 15:**

The system of Claim 1, wherein reasoning occurs within a **Persistent Existence Frame (PEF)** comprising:

- (a) a continuous present-state substrate,
- (b) reconstruction of prior conceptual states from echo-traces, and
- (c) projection of potential conceptual states.

Claim 16:

The system of Claim 15, wherein the PEF eliminates temporal encoding and sequence tracking.

Claim 17:

The system of Claim 15, wherein conceptual structures generated by primitives instantiate within the PEF and collapse back into the PEF upon completion.

Implementation Claims**Claim 18:**

The system of Claim 1, wherein the processing architecture is implemented using rule-based deterministic computation.

Claim 19:

The system of Claim 1, wherein primitive activation is implemented using constraint satisfaction algorithms.

Claim 20:

The system of Claim 1, wherein the architecture is implemented in hardware, software, or a hybrid system capable of executing structured conceptual operators.

Method Claims**Claim 21:**

A method for machine reasoning comprising:

- (a) maintaining a conceptual state of roles, domains, and spans,
- (b) evaluating primitive preconditions,
- (c) activating a compositional primitive,
- (d) applying update logic to modify the conceptual state, and
- (e) collapsing spans according to primitive-specific termination rules.

Claim 22:

The method of Claim 21, wherein ambiguity is preserved when multiple primitives or domain assignments are valid, and collapse occurs only when structural requirements force resolution.

Catch-all Claim**Claim 23:**

A machine reasoning architecture substantially as described herein with reference to any embodiment or combination of embodiments in this specification.

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